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Week 1 Summary

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1. Work Completed This Week

1.1. Development Tools Preparation

I started the week by preparing all the necessary software and tools with the help of IT Services. This includes Obsidian (to read Eugene's documentation), Typst (to compose documents), and Windows Subsystem for Linux (as a development runtime).

I have also submitted a request for access to the GPU clusters (a.k.a., Beatrix). I have set up a client for remote access and familiarized myself with the usage policies through the official documentation while the request is being reviewed.

1.2. Groundwork Study

I must point out that I have no previous experience with diffusion models. I need to build my knowledge from the ground up while working with the code. To achieve this, I scheduled 2 study sessions at work and 1 session at home per day, with each session being 90 minutes of focused learning.

Over the week, I have completed:

- **PyTorch Fundamentals**: Tensors, DataLoaders, and `nn.Module` architecture.
  Link to source
- **Transformer Architecture**: Self-attention mechanism, multi-head attention, and encoder-decoder structure
  Link to source

1.3. Preparing the Codebases

With some guidance from Eugene, I was able to install all the necessary runtime dependencies on my personal computer and the remote lab server. I then explored the two foundational projects: Motion Diffusion Model (MDM) and LoRA-MDM.

By running inference on MDM, I successfully generated human motion from text prompts. I also tested generation with different LoRA styles using the LoRA-MDM codebase.

Figure 1: Skeleton Animation Generated using Original LoRA-MDM Project

To better manage the project resources, I have organized the legacy code, literature, and documentation from Eugene into four Git repositories. All resources, as well as information about the project, are now stored securely on the NRC GitLab server, accessible to all researchers:

- **LoRA-MDM (Primary Codebase)**: *Gitlab Link*
- **VC Dataset Pipeline v2**: *Gitlab Link*

2. Regarding the Project

I want to share my honest assessment: given the amount of time left, **the project is behind schedule**. Based on Eugene's experiment journal, no publishable results have been generated so far. If this project is aiming for a submission on February 15th, ***it is at risk***.

I am proposing a shift in the project's focus.

2.1. Current Focus vs. Proposed Approach

2.1.1. Current: Continuous Age Conditioning

Core Idea	Treat age as a continuous variable (0.18–0.90) that smoothly controls motion characteristics
Technical	Modification of the MDM architecture; Custom `AgeEncoder` module injected into the diffusion pipeline
Validation	Velocity correlation justified (r=−0.96), but lacks visually observable difference
Status	Three experiments completed, but no publishable results; bug fixes ongoing
Risk	High — unproven innovation in model architecture
Timeline	Unknown — depends on results

2.1.2. Proposed: Discrete Age Groups

Core Idea	Use sparse age groups (Young, Adult, Elderly) as different motion "styles"
Technical	Standard LoRA adapters per group trained on Van Criekinge dataset; no architecture changes needed
Validation	Established LoRA-MDM metrics + age recognition user study
Status	To be conducted, but method proven on other styles ("Chicken", "Drunk", "Old")
Risk	Low — proven method, new domain
Timeline	4 weeks for training + validation; 3 more weeks for publication & deliverables

2.2. Justification

Eugene is now occupied with a busy school term, and I am still learning how the models work. This team setup is not suited for "reinventing the motion diffusion model architecture" in two months. While the idea of engineering a new model architecture is ambitious and intellectually interesting, it is **unlikely to generate publishable results by March**.

However, if planned properly, I am confident that we can produce a publishable result with the discrete age-group approach. This reframes the contribution as an **application-focused study** — demonstrating that LoRA adapters can capture age-related motion characteristics for assistive robotics — rather than a methods paper that improves the underlying model architecture.

This, of course, does not mean we are giving up on Eugene's plan — we are just delaying it. Continuous age control will be a future direction for this research. After the paper is accepted by a workshop, I will have more than a month for exploratory work on model architecture.

2.3. Proposed Timeline

Week 1	*(Completed)* Onboarding, foundational study, familiarizing with the project structure
Week 2	Complete foundational study; detailed review of existing research (MDM and LoRA-MDM) and their implementations
Week 3	Start working with the *Van Criekinge dataset* and its processing pipeline; familiarize with code implementation details
Week 4	**Fix all remaining issues** with the *Van Criekinge dataset*; split dataset by age group
Week 5	Train three distinct LoRA adapters: Young, Adult, and Elderly
Feb 15	Based on the rough results generated, **submit extended abstract**
Week 6–7	Fine-tuning; validation experiments; **detailed results and metrics**
Week 8–9	**Full paper writing**; additional experiments if needed
Week 10 Onwards	Exploratory experiments with new model architecture (Eugene's plan)

3. Week 2 Goals

1. Complete the foundational study on **diffusion models** (reverse process, training objective, sampling) and **transformers** (attention mechanism, encoders and decoders).
2. Read **MDM** and **LoRA-MDM** papers in detail.
3. Run exploratory experiments with the MDM and LoRA-MDM codebases for a more intuitive understanding.